

Smart Helmet for Proactive Rider Safety and Impact Detection

Abstract- *Two-wheeler riders face highly vulnerable to impacts and injuries, compounded by delayed emergency response times. Traditional helmets provide only surface protection without physiological monitoring or accident risk assessment. Our project presents a Smart Helmet integrating ECG, EEG, and IMU sensors with machine learning classification and sensor fusion techniques for proactive rider safety and impact detection. Independent models trained on publicly available datasets deliver risk-managed, explainable safety decisions. Experimental results demonstrate: ECG model accuracy of 93.02%, EEG fatigue detection rate of 90.7%, IMU crash detection rate of 82.7%, and fused system ROC-AUC of 100%. The integrated system enables both proactive accident prevention and emergency responder verification, making it ideal for intelligent wearable safety systems.*

Keywords- *Smart Helmet, ECG, EEG, IMU, Machine Learning, Sensor Fusion, Rider Safety.*

I. INTRODUCTION

Two-wheeler road traffic accidents are one of the major causes of global fatality, mainly in developing countries where motorcycles and scooters are commonly used [18]. Many regions consider helmets to be used as passive safety devices. They ignore causes of accidents, such as fatigue, cognitive stress, or sudden medical issues. Research shows that physiological and cognitive factors are crucial in accident occurrences, often occurring before any physical impact [6]. Recent advancements in wearable sensors, embedded systems, and deep learning based signal processing for biomedical usage allow the creation of smart safety systems that can monitor in real time and make decisions [17]. Restructuring a standard helmet into a smart device can greatly improve rider safety through proactive risk detection and initiating the emergency response process. Smart helmets with integration of bio signal monitoring have gained a lot of attention as potential real time usage in the near future [10]. The paper presents a multi-modal smart helmet system that integrates physiological monitoring, biosensing and motion analysis to enhance the rider's safety.

II. RELATED WORK

The most common smart helmet systems mainly focus on alerting users after accidents occur. Karthikeyan et al proposed an IoT-based smart helmet that makes use of GSM and GPS modules to send alert signals [1]. Ramesh et al. developed a system for helmet-to-vehicle communication that ensures helmets are worn before the vehicles are turned on [2]. These are useful for execution and these systems only react after incidents. Advanced

crash detection methods are having added specialized sensors. Zhou et al. used fibre Bragg grating sensors for impact detection with machine learning [3]. These systems are effective and increase hardware costs and complexity. Other studies like alcohol detection and threshold-based crash alerts [4], which are less flexible to different riding conditions.

Monitoring psychological states using machine learning has become very popular. Detection of cardiac abnormalities from ECG using deep learning has shown high accuracy [5]. Deep Convolutional Neural Networks is widely used as they have the ability to select temporal features directly from raw ECG signals [10]. Fatigue detection using EEG with wavelet analysis and machine learning has also produced good results [11]. EEG based cognitive state recognition performance is improved recently by graph-based neural architectures and attention-based models [9]. Multimodal fusion integrating EEG and ECG have improved robustness [7]. Making use of multimodal psychological systems improve the classification stability significantly in real world usage [8]. However, most existing systems either focus on a singular approach or the absence of a clear protection-oriented integration, making it suitable for real-time wearable deployment.

III. SYSTEM ARCHITECTURE

A. Overall System Design

Our proposed system has a portable design made up of sensing, preprocessing, analytics, integration, and notification. The block diagram of the smart helmet system is shown in Fig 1. All the sensors are working independently, and their outputs are merged at the final decision level to maintain separability and scalability.

B. Sensor Modalities

The ECG module continuously monitors the heart's electrocardiogram to detect abnormal conditions. The EEG module captures brain activity associated with fatigue and cognitive mental strain. The IMU module measures acceleration along three axes to detect collisions. Each sensor captures a variety of safety parameters, hence allowing continuous monitoring of the rider's safety.

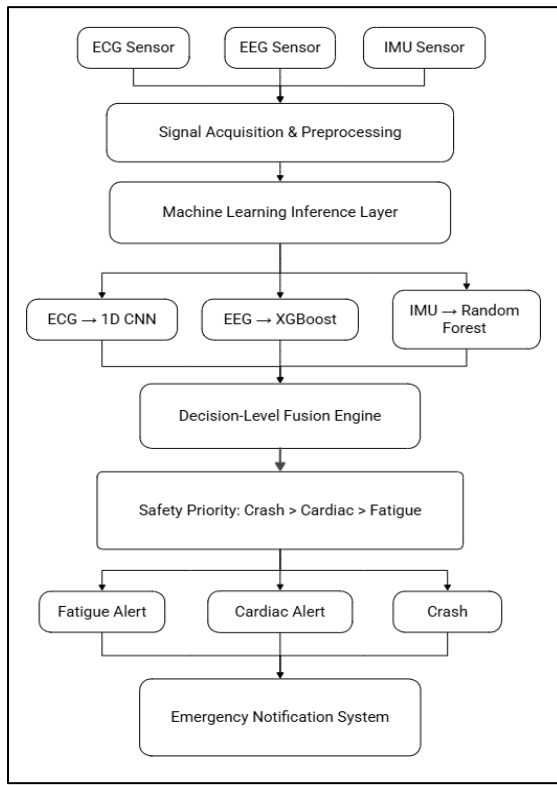


Fig. 1. Hardware-level block diagram of the proposed system

IV. METHODOLOGY

A. ECG-Based Cardiac Monitoring

ECG signals are taken from the MIT-BIH Arrhythmia Database with P-wave annotations [14]. Signals are sampled at 360 Hz and are divided into fixed-length windows (1s). This is trained using a 1D Convolutional Neural Network (CNN) on raw ECG windows to learn about the features related to P-wave activity. Convolutional Neural Network are widely used as they capture local patterns from raw ECG signals [10]. Its architecture is very effective in biomedical analysis related to time and also CNN is verified to perform well in ECG diagnosis tasks [19]. The model learns features directly from raw ECG windows, improving its generalization across unwanted signals.

B. EEG-Based Fatigue Detection

EEG-based fatigue detection uses the EEG Eye State Dataset [15], which has 14 frequency-domain and band-power features for the EEG channels. The XG Boost classifier is selected as it performs well on structured data and shows less signs of overfitting. XGBoost frameworks such as Gradient boosting are effective in structured physiological feature sets due to their normalization mechanism and scalable tree boosting formulation [13]. The classifier provides a probability score for all fatigue or normal cognitive states from the detection.

C. IMU-Based Crash Detection

IMU data collected from the MobiFall Dataset v2.0 [16] at a sampling frequency of 50 Hz. The recorded signals were separated into 1 second intervals where there was 50% overlap. Four extracted motion-related features, that are peak acceleration magnitude, mean acceleration, standard deviation, and peak jerk, are extracted from each window. A Random Forest classifier is trained to differentiate between normal daily activities and crash events. Random Forest classifier has high stability in recognising motion-based activity involving wearable inertial sensors [12].

D. Decision-Level Fusion

The probability results from ECG, EEG, and IMU models are combined. The hierarchy of combined results is set as follows: Crash detection (IMU), Cardiac abnormality detection (ECG), Fatigue detection (EEG). This makes sure that the system performs consistently and explains its decisions while keeping computational complexity low.

V. EXPERIMENTAL EVALUATION

A. Datasets

For the ECG MIT-BIH Arrhythmia Database with P-wave annotations (12 records, 264 annotations) [14], the EEG Eye State Dataset [16], and the IMU MobiFall Dataset v2.0 [17].

B. ECG Results

The ECG CNN model shows excellent generalisation, little overfitting, and stable performance despite of window lengths and sampling frequencies. The model achieves about 93.02% accuracy with an ROC-AUC of approximately 99.83% shown in Fig 2. The Overfitting analysis between train, test and validation is shown in Fig 3, also the results are, Sensitivity 86.41%, Specificity 99.63%, Precision 100%, F1-Score 0.9277.

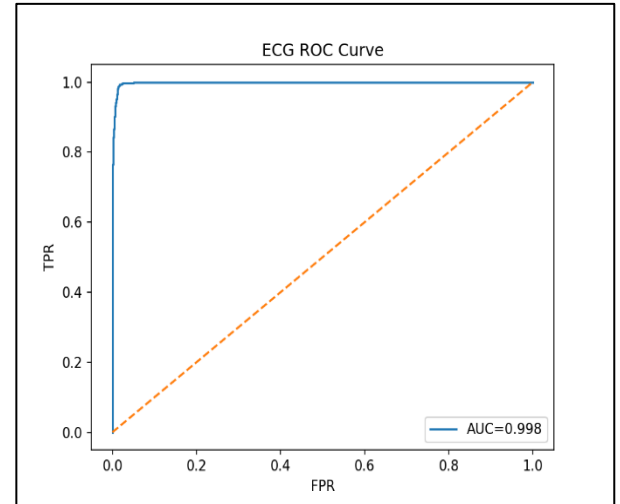


Fig 2: ECG CNN ROC Curve

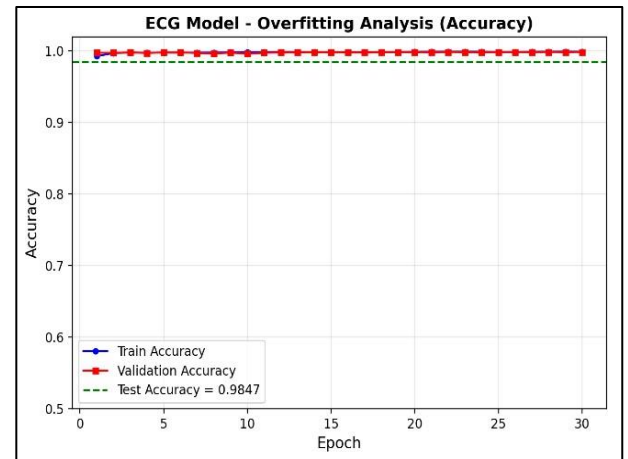


Fig 3: Overfitting Analysis – Train vs Test Accuracy

C. EEG Results

In Fig. 4, the confusion matrix shows effective class- wise discrimination, with the model accurately identifying most normal and fatigue states. A few misclassifications occur, which is expected given the noisy and subject-specific nature of EEG signals. The EEG fatigue detection model attains an accuracy of about 90.72%, with ROC–AUC values near 97.2% is shown in Fig 5 also the results of other parameters are Precision 93.11%, Recall 86.79%, Sensitivity: 86.41%

Specificity 99.63%, F1-Score 89.84%, and CV score $96.88\% \pm 0.27\%$. The impact of the number of EEG features on classification accuracy is shown in Fig.6. Results show a significant increase in accuracy as the number of features goes from 5 to 14, after which performance stabilises. This trend verifies the choice of 14 EEG features in the proposed system, balancing classification accuracy and computational efficiency. Using more features than 14 does not provide any significant change in results. The results of other parameters are Precision is 93.11%, Recall 86.79%, Specificity 94.24%, F1-Score 89.84%, and CV score $96.88\% \pm 0.27\%$.

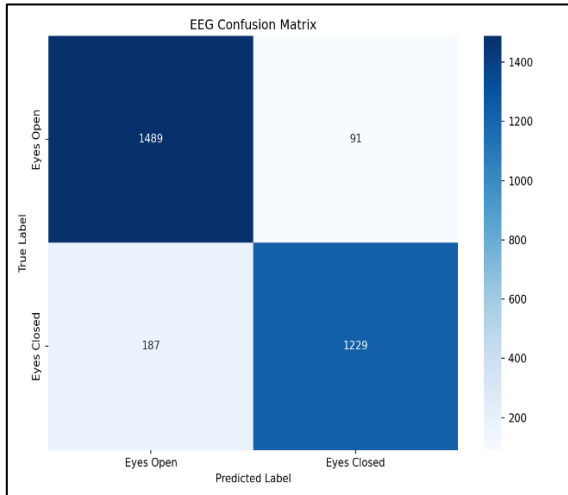


Fig 4: Confusion matrix of the EEG XGBoost Model

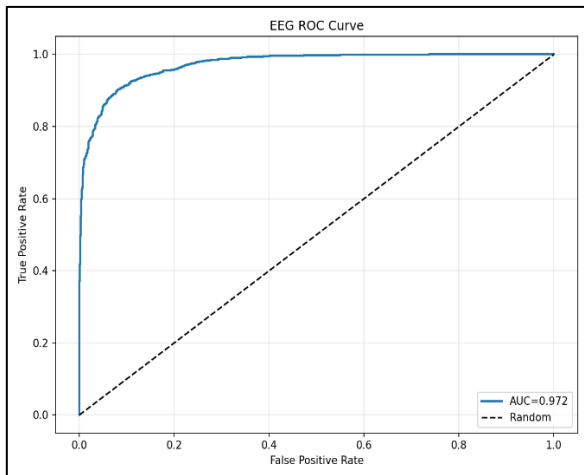


Fig 5: EEG XGBoost ROC Curve

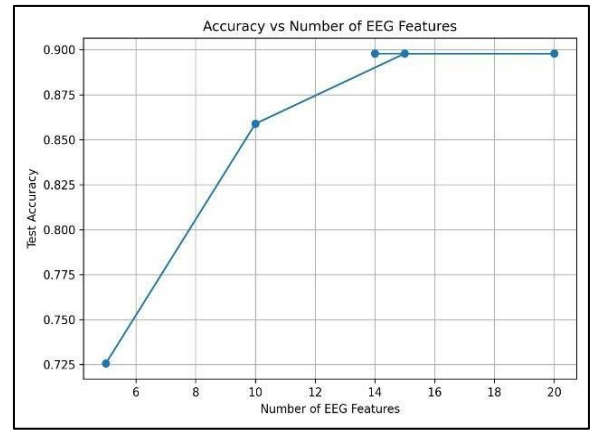


Fig 6: Accuracy vs Number of EEG Features

D. IMU Results

The IMU-based crash detection model has a consistent performance under varying conditions, achieving 82.70% accuracy with a test ROC–AUC around 89.7% also Sensitivity 77.96%, Specificity 84.11%. Collective learning approach have shown stable performance under noisy real-world conditions . This result is optimal to use in real time.

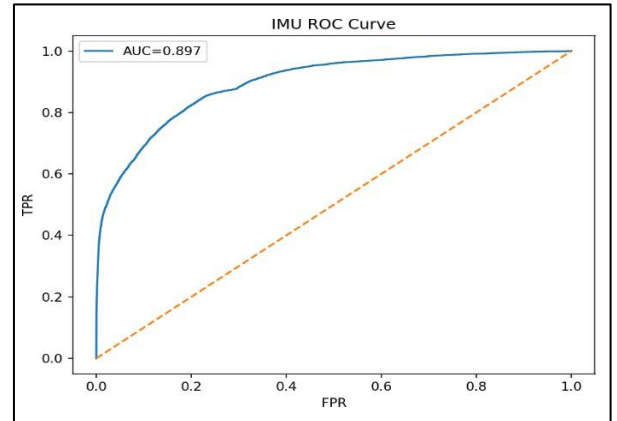


Fig 7: IMU Crash Detection ROC Curve (Test Data)

E. Fusion Results

The fusion system works perfectly on separate safety events, with ROC–AUC of 1.000, which shows the effectiveness of the safety-priority decision-level. Collective results are ECG 93.02%, EEG 90.72%, IMU 82.70% achieving an average performance of 88.81%.

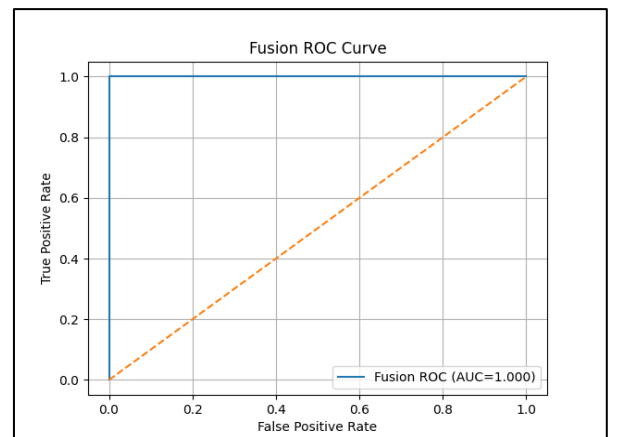


Fig 8: Fusion Model ROC Curve

VI. CONCLUSION & FUTURE WORK

A. Conclusion

Our paper presented a multi-modal smart helmet system that combines ECG, EEG, and IMU sensing with machine learning and decision-level

fusion to improve the safety of two-wheeler riders. The results of all the experiments show high accuracy, trustworthiness, and stability of ECG 93.02%, EEG 90.72%, IMU 82.70%, average 88.81% across all the methods used for sensing. The proposed multimodal decision level system is in track with edge-AI driven bio signal safety system. The fusion strategy for safety as priority shows clear and consistent behaviour, making the system suitable for real-world use.

B. Future Work

Future work will focus on building real-time embedded implementation and verify using live physiological data collected from riders. Further improvements will include integration of SpO₂ and respiration rate parameters. Additional Improvements will include large scale rider validation under real-world conditions to improve reliability and practical usage.

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